Regularity-Theoretic Inference Apparatus: Model Definitions and Proof Calculus

The Staged Inference Scheme Behind the Causal Intuitions Casebook

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Abstract

This document defines, as a family of formal inference apparatuses, all causal models developed within the regularity-theoretic tradition from Mackie's INUS analysis to the minimal theories of Graßhoff and May, the actual-causation theory of Baumgartner, the Boolean difference-making theory of Baumgartner and Falk, and the two regularity theories of Andreas and Günther. Each model is presented as a licence system for a staged inference: (i) normalisation of a natural-language situation into factors and variable assignments, (ii) construction of a coincidence table, and (iii) a derivation — a formal proof whose every step is justified by a named rule of the model. The rule calculus and the panel format defined here are used throughout the companion casebook (vol3-casebook.qmd).

keywords regularity theory · minimal theories · coincidence analysis · proof calculus · causal inference

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> **Note**
>
> **Causal Regularity Bundle — formal apparatus.** This document is an auxiliary component of the *Causal Regularity Bundle*. It supplies the formal model definitions and the proof calculus presupposed by the foundational paper (`arxiv-causal-regularity.qmd`, §"The deductive path to the causal hypothesis") and cited in every derivation of the casebook (`vol3-casebook.qmd`). Bundle concept and manifest: `bundle-hybrid-publication.qmd`, `bundle.yaml`.

1 Purpose and overview

The regularity-theoretic tradition has produced a sequence of causal models, each a refinement of its predecessors, and each definable as an *inference apparatus*: a set of rules that licenses the derivation of causal-relevance claims from non-causal data. This document fixes the apparatus family, the data format, and a uniform proof presentation, so that every case analysis in the casebook is a formal derivation whose steps can be checked mechanically.

Causal inference proceeds in four stages:

Stage I (**Normalisation**) translates a natural-language situation into a factor set with a variable assignment. Stage II (**Coincidence table**) records the empirically possible configurations of the factors — the Humean mosaic that constitutes the evidence. Stage III (**Derivation**) applies the rules of a selected model to the table, eliminating redundancies and checking instantiation and normality conditions, each step justified by a named rule. Stage IV (**Verdict**) states the resulting causal-relevance claims, to be confronted with the elicited intuition.

Throughout we use the crisp-set notation of coincidence analysis (Baumgartner and Falk 2023): italic capitals $A, B, C, \ldots$ denote factors (binary event types); A also denotes the value "present" and lowercase a the value "absent"; "$*$" is conjunction, "$+$" disjunction, "$\rightarrow$" material implication, "$\leftrightarrow$" material equivalence, and "$\Rightarrow$" a biconditional written in the causally oriented direction (antecedent = candidate causes).

2 Stage I: Normalisation

A case description delivers three items.

Factors. Each causally relevant event type mentioned or implied by the story is assigned a factor: a binary property that is either instantiated or not in each case of the domain. Factors must represent natural, modally independent properties: no factor may stand to another in a relation of logical or conceptual dependence, supervenience, constitution, or mereological containment (Baumgartner and Falk 2023, sec. 2; Baumgartner 2013, sec. 2). Normalisation makes the *causal field* explicit: the set of test situations recorded in the coincidence table, delimited by the factor set $\mathcal{F}$ and the population and time frame of the study (Mackie 1965; Graßhoff and May 2001). Separately, normalisation identifies the *background conditions* — the causally relevant factors of the process that are not elements of $\mathcal{F}$ and therefore unrecorded in the coincidence table. A factor varies within the field; a background condition is outside $\mathcal{F}$ entirely. No individual background condition is required to be constant. The constraint

on the background is bundle-level: for each latent cluster, the conjunction of its background literals must have constant truth-value — a condition satisfied by one commonly absent conjunct, not by demanding that every background factor be identical across all cases.

Factor set. The explicit, finite set $\mathcal{F} = \{A, B, \ldots\}$ of factors relative to which all sufficiency, necessity, and relevance claims are evaluated. Relativisation to $\mathcal{F}$ is essential: every model below quantifies over *suitable expansions* of $\mathcal{F}$ at some point.

Token assignment. For actual-causation questions, the configuration that the story instantiates: which factors are present, which absent, in the concrete episode. Where a story involves temporal structure that the factors do not resolve (late preemption), normalisation must introduce *marking factors* — factors whose instances distinguish process stages, such as "backup process running while the effect has not yet occurred" (Baumgartner 2013, sec. 3).

The output of Stage I is a normalisation table: one row per factor, with columns for the symbol, the type-level reading, and the token-level instantiation in the story.

3 Stage II: The coincidence table

The evidence for regularity-theoretic inference is a list of the empirically possible configurations of the factors in $\mathcal{F}$ — a *coincidence table* (in the terminology of coincidence analysis) or *Humean mosaic* δ over $\mathcal{F}$ (Baumgartner and Falk 2023, sec. 2). Each row is a type of co-instantiation; the table records which types exist in the domain. Two properties matter:

Homogeneity. This is the precondition on which the whole regularity-theoretic inference rests, and it is due to Graßhoff and May (2001) with the detailed treatment in Baumgartner and Graßhoff (2004); we state it explicitly. *Canonical definition: register sense II.1 (bundle-level minimal homogeneity, operative for sufficiency-reading); sense III.1 (contrastive bundle homogeneity, operative for difference-tests). Sense I (†HC-strong) is documented and rejected. See* `paper/register/causal-regularity.qmd`, *§"Homogeneity Family".*

> **Homogeneity condition (principle of the homogeneous background).** A coincidence table δ over a factor set $\mathcal{F}$ is *homogeneous* with respect to an outcome E iff there is no pair of rows that agree on the values of all factors in $\mathcal{F} \setminus \{E\}$ yet disagree on the value of E. Equivalently: every configuration of the non-outcome factors that is observed with E present is never also observed with E absent. A pair violating this — a *contradictory configuration* — establishes that some difference-maker of E lies outside $\mathcal{F}$: the observed factors do not fix E, so an unmeasured factor must vary between the two rows.

The statement above is the *operational* form of the condition — directly checkable on the table. Its *definitional* form concerns the causal background: for every minimally sufficient condition M_j of E in the complete structure over measured and latent factors alike, the *truth-value of the background bundle* L_j — the conjunction of M_j's latent conjuncts — is constant across the test situations: either L_j always fires, or never fires. This is a condition on the bundle's logical value, not on the identical

instantiation of each latent factor individually. Individual factors inside L_j may vary freely from case to case — including different blocking conjuncts in different cases — as long as the bundle's overall truth-value does not change. The condition is most cheaply met by a single shared absent conjunct per latent cluster (which holds L_j at false regardless of all other conjuncts), and every other latent factor may vary freely. This is *minimal* homogeneity, and it is deliberately far weaker than the totality clause inherited from Mill — that every factor causally relevant to E and outside $\mathcal{F}$ be constant — which the foundational paper states as (HC-strong) and rejects, since no real field satisfies it and a condition that strong would prohibit causal inference rather than license it (`arxiv-causal-regularity.qmd`, §"Mill's error: the totality formulation"). Given the principle of causal determination, the definitional form entails the operational one, so a contradictory configuration refutes background homogeneity outright, and with it the deductive schema the two forms underwrite: given homogeneity and causal determination, the validity of a causal hypothesis follows deductively from the coincidence table by the rules of Section 5.

The condition is what licenses reading sufficiency and necessity off the table at all: without it, a configuration's association with E carries no information, because the same configuration also occurs without E. Graßhoff and May (2001) make homogeneity of the causal background the standing presupposition of every minimal-theory inference — all sufficiency and necessity claims are relativised to a *homogeneous causal field* — and Baumgartner and Graßhoff (2004) give the systematic account of how contradictory configurations are to be diagnosed and repaired. The repair is not to discard the data but to *expand* $\mathcal{F}$ by the latent factor the contradiction reveals (rule EXPAND / permanence, Section 5): homogeneity restored by expansion is exactly how the epiphenomenon and the LK-99 cases are solved. A contradictory configuration is therefore not noise but the theory's built-in detector of confounding.

Completeness. A table is complete for the underlying structure iff it lists all configurations the structure permits. Completeness cannot be verified from the table itself — this is why every model since Baumgartner (2013) makes causal interpretability conditional on *permanence* across suitable expansions, and why inference from incomplete tables is inductive (Baumgartner and Falk 2023, sec. 7).

For the epistemic apparatus of Andreas and Günther (2019) (model **RT-B** below), Stage II is replaced by a prioritised belief base: generalisations and particular facts ordered by entrenchment. The belief base plays the role of the mosaic; contraction and expansion play the role of row selection.

4 The models

The family comprises nine apparatuses, eight of them causal models. The ninth, RT-B, is an epistemic apparatus and is included for its role as the original of three rules; it is therefore absent from Table 2, which compares causal models. Each is stated with its licensed rules (Section 5); later models inherit the structural core of earlier ones.

4.1 INUS – Mackie 1965/1974 (historical baseline)

A is a cause of E (in field F) iff A is at least an INUS condition of E: an insufficient but non-redundant part of an unnecessary but sufficient condition (Mackie 1965, 1974). Formally: there are X, Y such that $A * X + Y \leftrightarrow E$, where $A * X$ is minimally sufficient for E (X possibly empty), and the disjunction is necessary but *not required to be minimally* necessary.

Rules: (SUF), (MS), (NEC). **Known defect:** because necessity is not minimised, INUS overgenerates on common-cause structures. Mackie himself pointed out that on his analysis the sounding of the Manchester factory hooters counts as a cause of the London workers' going home (a case adapted from Russell (Andreas and Günther 2026, sec. 1.3)), and, seeing no repair within regularities, abandoned the purely regularity-theoretic analysis of causation (Mackie 1974, 81–87; Baumgartner 2008, sec. 7). INUS is included here as the baseline against which the minimal theory is defined; it is used only for contrast.

4.2 MT – the minimal theory (Graßhoff & May 2001)

An *atomic minimal theory* of E over $\mathcal{F}$ is a biconditional

$$\Sigma_1 + \Sigma_2 + \ldots + \Sigma_k \Rightarrow E$$

such that each Σ_i is a minimally sufficient condition of E (no conjunct eliminable without loss of sufficiency), the disjunction is minimally necessary (no disjunct eliminable without loss of necessity, so that in particular no Σ_i subsumes another), membership is stable under suitable expansions of $\mathcal{F}$ (permanence), and the biconditional is causally oriented by the asymmetry of determination: each Σ_i determines E, but E does not determine which Σ_i is instantiated — which requires $k \geq 2$ (Graßhoff and May 2001; Baumgartner 2008, definition (VIII)). A is *causally relevant* to E iff A or a occurs as a literal in some Σ_i of a minimal theory of E.

The condition of minimal necessity is the historically decisive addition: it is what first solved the Manchester Factory Hooters problem — the problem Mackie had introduced in 1974 to show the limits of INUS and had himself declared irreparable within regularities. The solution is due to Graßhoff and May (2001) and, synchronously, May (1999); Baumgartner and Graßhoff (2004, 99–103) give the detailed treatment, and Baumgartner (2008, sec. 7) the systematic generalisation to all epiphenomenon structures. The worked exemplar in Section 6.1 displays the solution as a derivation.

Rules: (SUF), (MS), (NEC), (MN), (INST), (CPLX), (PERM). Token-level supplement (Baumgartner 2008, sec. 9): a causes b iff their types are MT-related and every other conjunct of the relevant Σ_i is instantiated coincidently with a — the germ of the active cluster.

4.3 TC/AC – type and actual causation (Baumgartner 2013)

(TC) A is a type-level cause of E iff there is a factor set $\mathcal{F}_i \ni A, E$ such that A is part of a minimal theory of E over $\mathcal{F}_i$, and for every suitable expansion $\mathcal{F}_j$ there is a minimal theory of E over $\mathcal{F}_j$ containing A — in short, iff A is permanently non-redundant for E (Baumgartner 2013, sec. 2).

(AC) Token a is an actual cause of token e iff (i) A is a type-level cause of E by (TC), and (ii) relative to every suitable expansion, A is connected to E by an *active cluster* (Baumgartner's *active causal route*) on which a and e are the instances: a chain of direct minimal theories in each of which the

antecedent factor is co-instantiated with all remaining conjuncts of its minimally sufficient condition (Baumgartner 2013, sec. 3).

Rules: MT rules plus (AR), (MARK), (PERM) extended to active clusters.

4.4 cAC – contextualised actual causation (Baumgartner 2013, §4)

A pragmatic supplement, explicitly no part of the core theory: given a context-supplied *typicality ranking* of configurations, delete all rows of the coincidence table ranked less typical than the actual configuration and re-minimise. a is a contextualised actual cause of e iff it satisfies (AC) and its factor is part of the resulting *contextually weighted* minimal theory (Baumgartner 2013, sec. 4).

Rules: AC rules plus (TYP). cAC is the apparatus that replicates bogus-prevention and default-omission intuitions *if* a typicality ranking is supplied; the core verdict remains available alongside.

4.5 CMT – complex minimal theories (multi-outcome structures)

A *complex minimal theory* is a conjunction $\Phi_1 * \ldots * \Phi_n$ of atomic minimal theories with distinct outcomes, representing chains, common-cause structures, and cycles. Two conditions govern the conjunction (Baumgartner and Falk 2023, sec. 6): (a) *structural minimality* — the conjunction contains no conjunct eliminable without loss of logical content relative to the mosaic; (b) *distinct consequents* — no two conjuncts share an outcome (ambiguous mosaics are represented by disjunctions of complex minimal theories, not by merged ones).

Rules: MT rules plus (SM), (DC). CMT is the data structure implemented in Causal-Architect as `ComplexMinimalTheory` (csf), with atomic minimal theories as `AtomicMinimalTheory` (asf).

4.6 CR – Boolean difference-making (Baumgartner & Falk 2023)

The completion of the type-level programme. Let Ω be a Boolean expression in DNF. **(CR)**: Ω is causally relevant to B iff there is a factor set $\mathcal{F}_\delta$ containing B and all factors in Ω, with mosaic δ, such that (a) there is an atomic minimal theory $\Pi \Rightarrow B$ for δ — an RDN-biconditional contained in a structurally minimal conjunction — with Ω contained in Π, and (b) the same holds for every suitable expansion of $\mathcal{F}_\delta$: containment in a *permanently redundancy-free atomic minimal theory* (Baumgartner and Falk 2023, sec. 6). CR is underwritten by the non-redundancy principle (NR) and the metaphysical assumption of causal uniqueness (CU): every complete mosaic corresponds to exactly one determinate causal structure.

Rules: CMT rules plus (RDN), (PERM) at structure level. CR-relevance is non-transitive by construction (the switch case).

4.7 AG-R – reductive regularity theory (Andreas & Günther 2024b)

A causal model is a pair $\langle L, F\rangle$: L a set of *direct non-redundant regularities* (RDN-biconditionals additionally required to be direct), F the true particular facts. A *setting* $[\cdot]$ adds a proposition after removing the regularity of its variable, cutting off backward inference. c is a cause of e iff there are $F' \subseteq F$ and $L' \subseteq L$ such that

(1) $\langle L, F\rangle \models C \wedge E$;
(2) $\langle L', F'\rangle \not\models E$, and no F'' with $F' \subset F'' \subseteq F$ satisfies $\langle L', F''\rangle \not\models E$ (maximality);
(3) $\langle L', \varnothing\rangle[F'][\{C\}] \models E$ (forward-directed inferability);

(4) every $F^{\dagger} \in F \setminus F'$ that is $\equiv C$ or a root proposition is more deviant than its negation (deviancy);
(5) the regularity of every descendant of C is in L' (path condition) (Andreas and Günther 2024b).

Deviancy conventions: an occurrence is prima facie more deviant than its absence (Gallow 2021); an absence is more deviant than the occurrence if it violates a norm active in the scenario (McGrath 2005).

Rules: (SUF), (MN)-inherited via L, (MAX), (SET), (LREL), (PATH), (DEV).

4.8 AG-L – Lewisian regularity theory (Andreas & Günther 2024a)

The non-reductive sequel: L contains directed *law-like propositions* $A = \phi$ (direction primitive), and the deviancy condition is revised — forced by the assassin's-mentor case — to: (v) every $C' \in F \setminus F'$ whose variable is neither a descendant nor an ancestor of C is more deviant than $\neg C'$ (Andreas and Günther 2024a, definition 2). Slogan for both AG models: causation is deviant forward-directed inferability along lawful paths.

Rules: as AG-R, with (DEV′) replacing (DEV).

4.9 RT-B – strengthened Ramsey test for 'because' (Andreas & Günther 2019)

The epistemic ancestor. Relative to a prioritised belief base H with epistemic state $K(H, <)$: $\alpha \gg \gamma$ iff after suspending judgement about α and γ (contraction by $B(\alpha) \vee B(\gamma)$), adding α makes γ inferable. "Because α, γ" iff $\alpha, \gamma \in K$ and $\alpha \underset{s}{\gg} \gamma$, where $\underset{s}{\gg}$ permits prior contraction of the base by a set of literals (ignoring parallel explanatory paths) (Andreas and Günther 2019, secs. 6–7). Not a causal model; included because its operations (agnostic contraction, directed inference tickets, contractible backgrounds) are the epistemic originals of (SET), (MAX), and (LREL).

Rules: (SUSP), (ADD), (CONTR).

5 Stage III: the proof calculus

Every derivation step in a casebook panel cites one of the following rules. Rules are stated as licences: what may be inferred or eliminated, on what grounds, relative to which model. The methodological point of the calculus deserves emphasis (Baumgartner and Graßhoff 2004): relative to the premisses of Stage II — the coincidence table, its homogeneity, and the principle of causal determination — every derivation is *deductively valid*. Inductive risk is carried entirely by the premisses, never by the steps (Figure 1).

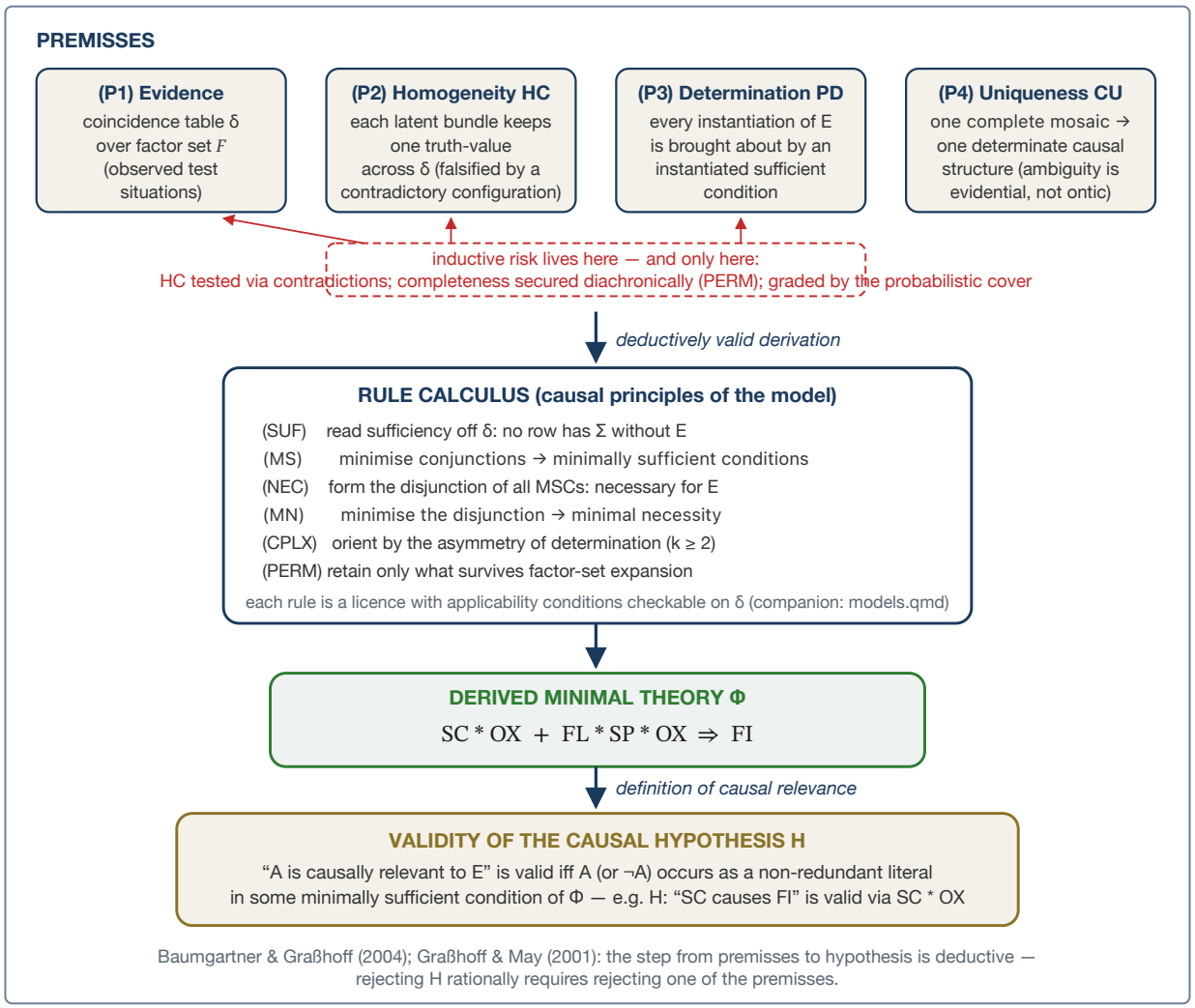


Figure 1 The deductive path from assumptions and causal principles to the validity of the causal hypothesis. The rule calculus of this section performs the middle step; the premisses carry the inductive risk.

Table 1 The rule calculus

Rule	Licence	Models
(SUF)	Read off $\Sigma \rightarrow E$ from the table: no row has Σ without E	all
(MS)	Eliminate conjunct X from sufficient $\Sigma * X$: Σ alone sufficient (includes elimination of trivially true conjuncts)	all from MT
(NEC)	The disjunction of all minimally sufficient conditions is necessary for E	all

Rule	Licence	Models
(MN)	Eliminate disjunct Σ_i from a necessary condition: remainder still necessary	MT, TC/AC, CMT, CR
(INST)	Every disjunct of a minimally necessary condition is instantiated together with E when all other disjuncts are absent (May's existential implication)	MT and successors
(CPLX)	Causal orientation: (a) ≥ 2 alternative MSCs — asymmetry of determination settles the direction; (b) $k = 1$ with an explicit Stage-I process order — temporal precedence orients the single-cluster biconditional as `order-oriented`	MT and successors
(RDN)	Form the RDN-biconditional $\Pi \Rightarrow B$: minimally necessary disjunction of minimally sufficient conditions, DNF	CR
(SM)	Eliminate a whole biconditional that is structurally redundant: the remaining conjunction is logically equivalent relative to δ	CMT, CR
(DC)	Split biconditionals with identical consequents into alternative minimal theories	CMT, CR
(PERM)	Retain a component only if no suitable expansion of $\mathcal{F}$ eliminates it; spurious components are unmasked by expansion	TC/AC, CR
(AR)	Token verdict: candidate lies on an active cluster — co-instantiated with all remaining conjuncts of its minimally sufficient condition, chained through direct minimal theories	AC

Rule	Licence	Models
(MARK)	Introduce a marking factor to resolve process stages	AC
(TYP)	Truncate rows less typical than the actual configuration; re-minimise	cAC
(MAX)	Fix background F': maximal subset of facts from which E is not inferable	AG-R, AG-L
(SET)	Setting: add proposition after removing its variable's regularity; inference runs forward only	AG-R, AG-L
(LREL)	Relax the regularity set to $L' \subseteq L$ (entangled causes)	AG-R, AG-L
(PATH)	The law of every descendant of the candidate must be retained in L'	AG-R, AG-L
(DEV)	Deviancy check on root/off-path facts outside F' (AG-R: root propositions; AG-L: neither-descendant-nor-ancestor variables)	AG-R, AG-L
(SUSP)	Suspend judgement: contract by $B(\alpha) \vee B(\gamma)$	RT-B
(ADD)	Hypothetically add the antecedent to the contracted base	RT-B
(CONTR)	Contract the base by a set of literals (ignore parallel paths)	RT-B

A **derivation** is a numbered sequence of lines. Each line consists of a claim (a Boolean formula over $\mathcal{F}$, a model-theoretic statement, or an elimination) and a justification: the rule applied and the lines or table rows it operates on. A derivation is *closed* when it terminates in a minimal theory (type-level question) or a cause/non-cause verdict for the token assignment (actual-causation question).

6 The inference panel

Casebook analyses are presented in a uniform five-part *inference panel*, set off from the narrative text (in the web presentation as a tinted, rule-bordered block; in print as an indented block):

I Normalisation — the variable table: symbol, factor (type-level reading), token instantiation. **G Causal graph** — a CVL diagram of the structure under test, with the token layer shaded. CVL, defined in the companion proposal `causal-visuals.qmd`, is isomorphic to the formal representation: factor

nodes, conjunction arcs drawn across the arrows of the jointly required parts of one complex cause, a single '¬' box on the shaft for a negated factor, hyperedge in-bundles as minimally necessary disjunctions, and module boxes with exact collapse/expand semantics. **Rule (Stage G): every case receives a causal graph.** Derivation steps are graph transformations — (MS) deletes one arrow from a cluster, (MN) deletes a whole cluster (a single arrow or an arc-tied group of arrows), (SM) deletes an in-bundle, (AR) thickens the active cluster. Converters between the formal and the visual representation (both directions, round-trip tested) are provided in `casebook-sources/cvl.py`. **II Coincidence table** — the mosaic δ; the token row is marked "◀" where an actual-causation question is asked. **III Model** — the apparatus applied (badge: MT, AC, cAC, CR, AG-R, AG-L, RT-B), with divergent apparatuses given parallel derivations. **IV Derivation** — the proof table. **V Verdict** — the relevance claims, confronted with the intuition.

6.1 Worked exemplar: Manchester Factory Hooters

The exemplar is chosen deliberately: this is the case Mackie constructed to show that his INUS analysis certifies a non-cause, and the case that Graßhoff and May (2001) were the first to solve — by the minimisation of necessary conditions, rule (MN). The derivation runs the INUS apparatus to its stopping point (line 4) and then completes the minimisation (lines 5–8), so that the solution is visible as exactly one additional rule application. The casebook presents the same derivation with a full plain-language commentary column as the *canonical derivation*; all case panels follow its pattern.

I Normalisation.

Symbol	Factor (type)	Token
T	it is five o'clock (knocking-off time)	instantiated
D	abnormal condition triggering the Manchester hooters off-schedule	absent
B	alternative cause of the Londoners' going home	absent
M	Manchester factory hooters sound	instantiated
L	London workers go home	instantiated

Causal field: the auxiliary standing conditions C_2, C_3 of Mackie's formulation (functioning hooter mechanisms, working Londoners) are held fixed. $\mathcal{F} = \{T, D, B, M, L\}$; structure under test: $T + D \leftrightarrow M$, $T + B \leftrightarrow L$.

G Causal graph.

II Coincidence table. All empirically possible configurations:

T	D	B	M	L
0	0	0	0	0
0	0	1	0	1
0	1	0	1	0

T	D	B	M	L	
0	1	1	1	1	
1	0	0	1	1	◀
1	0	1	1	1	
1	1	0	1	1	
1	1	1	1	1	

III Model. MT (the INUS baseline is run first, to exhibit the defect).

IV Derivation. Outcome factor: L.

#	Claim	Justification
1	$T \to L$; $B \to L$; $M * \neg D \to L$	(SUF), rows 5–8; 2,4,6,8; 5,6
2	T, B, $M * \neg D$ are minimally sufficient (M alone: row 3 refutes; $\neg D$ alone: row 1 refutes)	(MS)
3	$T + B + M * \neg D \leftrightarrow L$	(NEC), rows 2,4–8
4	*INUS verdict:* M is an INUS condition of L — the hooters cause the Londoners to go home	INUS stops at line 3
5	$T + B$ is necessary for L: every L-row (2,4,5,6,7,8) contains T or B	(NEC)
6	Eliminate $M * \neg D$: $T + B$ is a necessary proper part of line 3	(MN)
7	Neither T nor B alone necessary (row 2 lacks T; row 5 lacks B); $k = 2 \geq 2$	(MN), (CPLX)
8	Minimal theory of L: $T + B \Rightarrow L$	lines 2, 6, 7: (MS), (MN), (CPLX)

V Verdict. $M \notin$ any minimally sufficient condition of the minimal theory of L: the sounding of the Manchester hooters is not causally relevant to the Londoners' going home. The MT verdict replicates the intuition; the derivation exhibits the exact point (line 6) at which Mackie's apparatus, lacking

(MN), is forced into the counterintuitive verdict of line 4 — and at which Graßhoff and May's minimisation of necessary conditions first solved the problem Mackie had judged insoluble.

7 Model comparison

Table 2 Conditions by model

Condition	INUS	MT	TC/ AC	cAC	CMT	CR	AG-R	AG-L
Minimal sufficiency	✓	✓	✓	✓	✓	✓	via L	via L
Minimal necessity	—	✓	✓	✓	✓	✓	via L	via L
Direction from asymmetry of determination	—	✓	✓	✓	✓	✓	—	primitive
Permanence under expansion	—	✓	✓	✓	✓	✓	(stability desideratum)	—
Structural minimality	—	—	—	—	✓	✓	directness of L	—
Token instantiation	—	(§9 suppl.)	active cluster	active cluster	—	—	maximal F' + setting	maximal F' + setting
Path condition	—	—	—	—	—	—	✓	✓
Normality	—	—	—	typicality (pragmatic)	—	—	deviancy (in definition)	revised deviancy
Reductive	✓	✓	✓	✓	✓	✓	✓ (ambition)	×
Level	type	type	type + token	token	type	type	token	token

The table makes the casebook's stratification result visible in the apparatus itself: the structural columns (rows 1–5) are settled and shared; the instantiation machinery exists in two intertranslatable variants (active clusters; maximal backgrounds with settings); normality enters either pragmatically (cAC) or definitionally (AG), and nowhere else does any model diverge from any other.

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